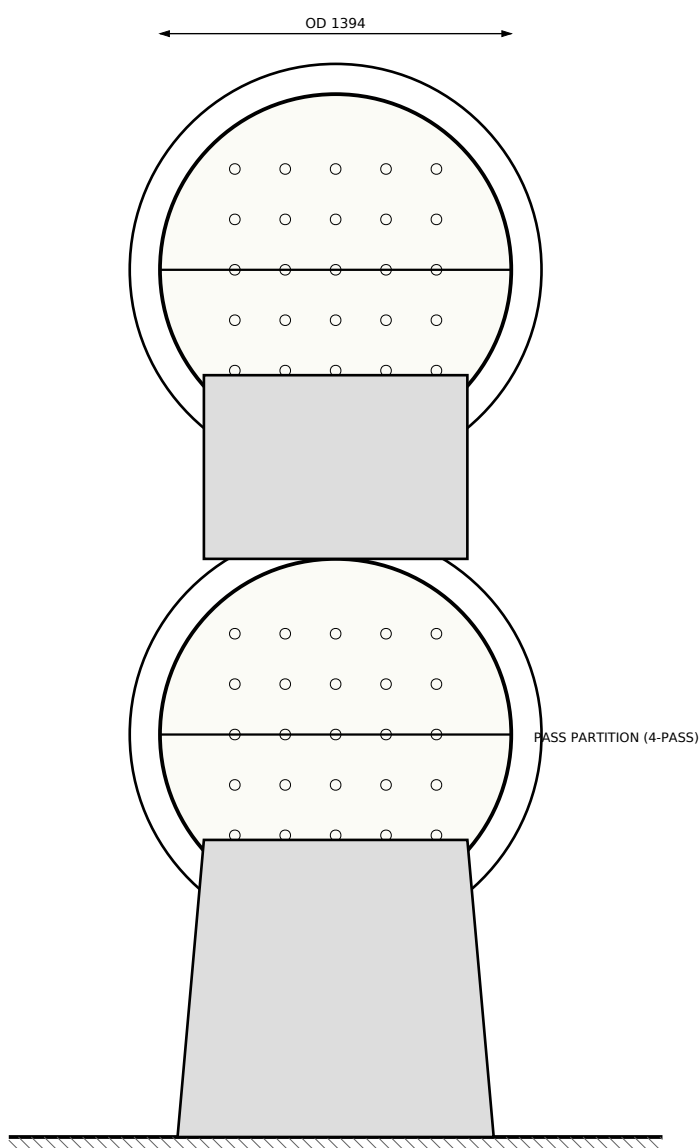


DESIGN DATA	
Item	E-106 - Crude / Middle (kero) pumparound
TEMA type / size	1350-6096 AES (Ds-L mm)
Shells	2 (2 series x 1 parallel), stacked 2-high
Surface	500 m ² /shell; 1001 m ² total
Duty / U	12.8 MW / 320 W/m ² K
Tubes	1055 x 25.4 OD 12 BWG (2.77 mm), 6096 mm, 4 passes
Pitch	31.75 mm square (cleanable)
Design P shell / tube	30 / 20 barg
Design T	270 C
Shell matl. / t	SA-516-70 / 22 mm
Tubes matl.	CS shell / 5Cr-1/2Mo tubes
Weight empty / filled	25.2 / 35.6 t per shell
Code	ASME VIII-2 / TEMA R / API 660

END VIEW (CHANNEL END) 1:30

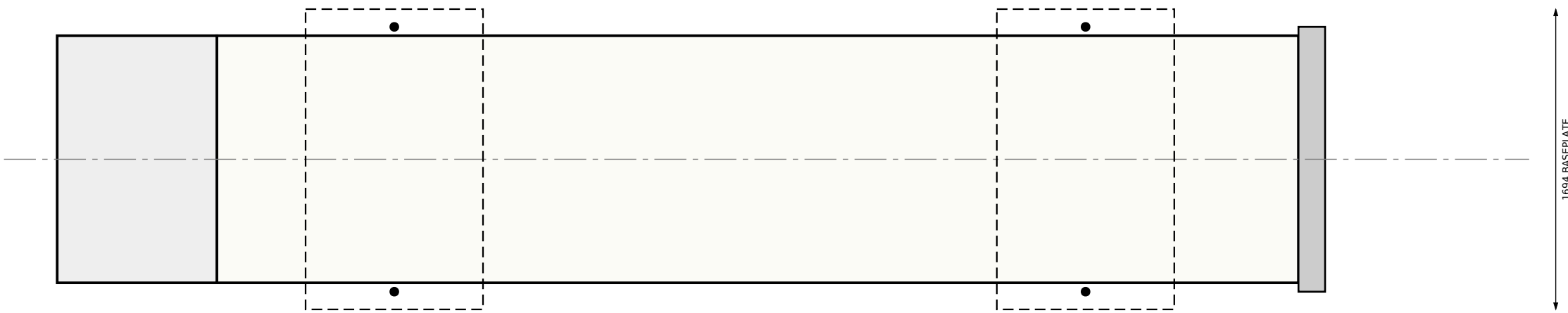
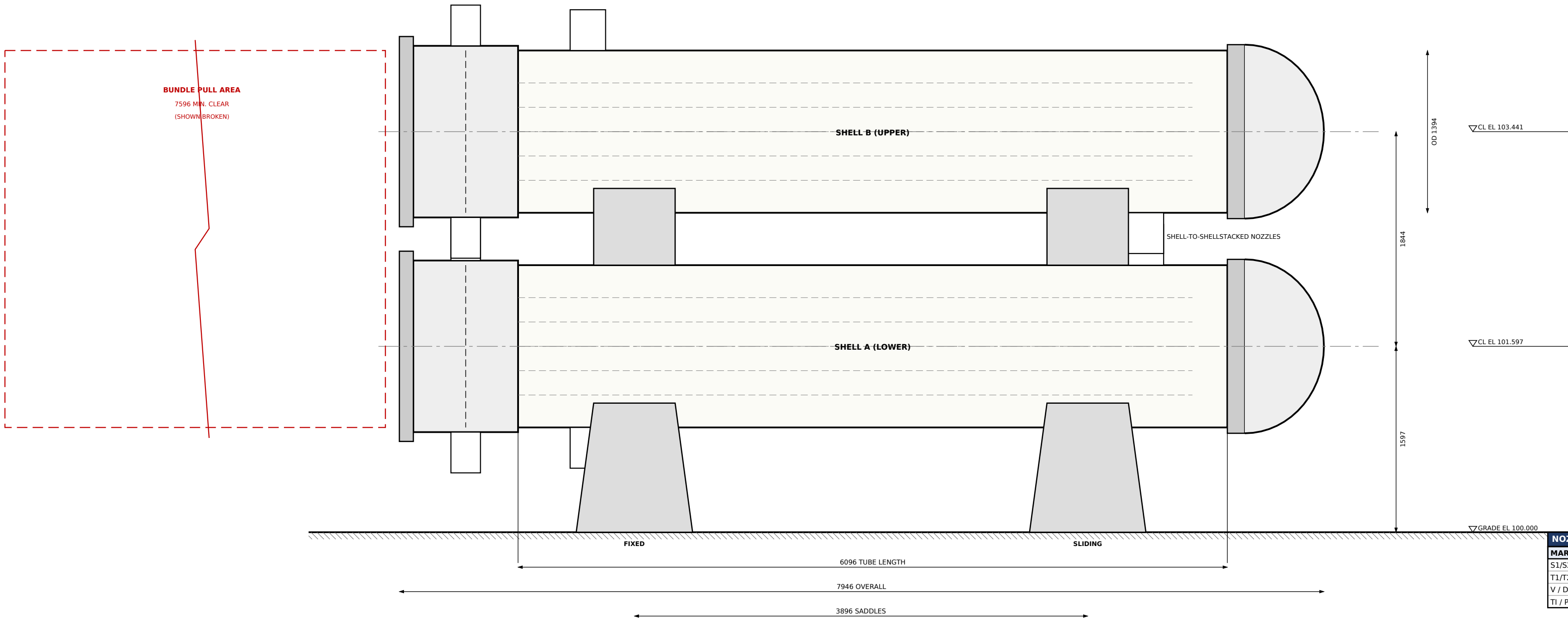


NOZZLE SCHEDULE (EACH SHELL)					
MARK	QTY	SIZE	RATING	SERVICE	
S1/S2	2	12"	300# RF	Shell in / out (crude)	
T1/T2	2	10"	300# RF	Channel in / out (MPA)	
V / D	4	1"	300# RF	Vents / drains (each shell)	
T1 / P1	4	1"	300# RF	Thermowell / pressure connections	

AES SHELL SCHEDULE (TYPICAL ARRANGEMENT APPLIES)							
TAG	SIZE	ARRGT	m ² /SH	Ds mm	t mm	S/SH	SERVICE
E-101	1200-6096 AES	15 x 1P	359	1200	20	18.9	Crude / Top pumparound
E-102	1300-6096 AES	15 x 1P	438	1300	20	22.4	Crude / Kerosene product
E-103	1250-6096 AES	25 x 1P	402	1250	20	20.7	Crude / LVGO pumparound/product
E-104	1200-6096 AES	15 x 1P	378	1200	20	19.5	Crude / Diesel product
E-105	1300-6096 AES	15 x 1P	431	1300	22	22.7	Crude / Vacuum residue
E-106	1350-6096 AES	25 x 1P	500	1350	22	25.2	Crude / Middle (kero) pumparou
E-107	1150-6096 AES	15 x 1P	324	1150	22	18.2	Crude / Diesel product
E-108	1400-6096 AES	25 x 2P	549	1400	25	28.2	Crude / HVGO pumparound/product
E-109	1000-6096 AES	15 x 1P	263	1000	20	14.5	Crude / AGO product
E-110	1300-6096 AES	15 x 2P	443	1300	22	23.2	Crude / Bottom (diesel) pumpar
E-111	1400-6096 AES	15 x 2P	547	1400	25	28.3	Crude / Vacuum residue
E-114	650-6096 AES	15 x 1P	86	650	10	5.0	Stabiliser feed / bottoms

ELEVATION 1:30

PLAN 1:30



ANCHOR BOLTS 4 x M24 PER SADDLE (SLOTTED AT SLIDING SADDLE)

- NOTES**
- Dimensions in mm; elevations in m (grade EL 100.000).
 - Typical for AES crude preheat shells E-101, E-111; E-106 shown (2 shells in series, stacked).
 - Crude shell side (square pitch, removable bundle); hot stream tube side.
 - Bundle-pull area = tube length + 1.5 m clear at channel end.
 - Lower shell saddle fixed, upper shells sliding; intermediate supports bolted.
 - Datasheet CFU-000-ME-DS-004; weights data/mech.json.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
EXCHANGER SETTING PLAN TYPICAL CRUDE PREHEAT EXCHANGER - TEMA AES (E-106)		
DISCIPLINE	STAGE	SCALE
MECHANICAL	FEED	1:30
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DRAWING NUMBER	REV	
CFU-100-ME-GA-008	A	